

AI/ML intern DAY 16

Body Measurement Extraction Research Report

(Member-1)

LOYA PRANAY (pranay8679@gmail.com)

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1. Introduction

Body Measurement Extraction is a computer vision application that estimates human body dimensions from a single image using Artificial Intelligence and pose estimation techniques.

The technology is widely used in:

- Virtual Try-On systems
- Fashion recommendation platforms
- Online clothing stores
- Fitness tracking applications
- Digital tailoring systems
- Avatar generation

Traditional body measurement methods require manual measurement tools, whereas AI-based approaches use body landmarks and computer vision algorithms to estimate body dimensions automatically.

This research focuses on studying body measurement extraction using MediaPipe Pose and OpenCV.

2. Objectives

The objectives of this research are:

- Understand body measurement extraction techniques
- Study body landmark detection
- Estimate body dimensions from images
- Analyze measurement accuracy
- Evaluate suitability for Virtual Try-On systems
- Identify challenges and limitations

3. Body Measurement Extraction Workflow

The measurement extraction process follows:

Input Image

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Human Detection

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Pose Estimation

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Landmark Detection

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Measurement Calculation

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Output Generation

The detected body landmarks are used to estimate various body measurements.

4. MediaPipe Pose for Measurement Extraction

Overview

MediaPipe Pose is a pose estimation framework developed by Google that detects 33 body landmarks.

The detected landmarks provide coordinate information that can be used for body measurement estimation.

Important Landmarks

- Left Shoulder
- Right Shoulder
- Left Hip
- Right Hip
- Left Elbow
- Right Elbow
- Left Wrist
- Right Wrist
- Knees
- Ankles

Applications

- Height estimation
- Shoulder width estimation
- Waist measurement
- Hip measurement
- Arm length estimation

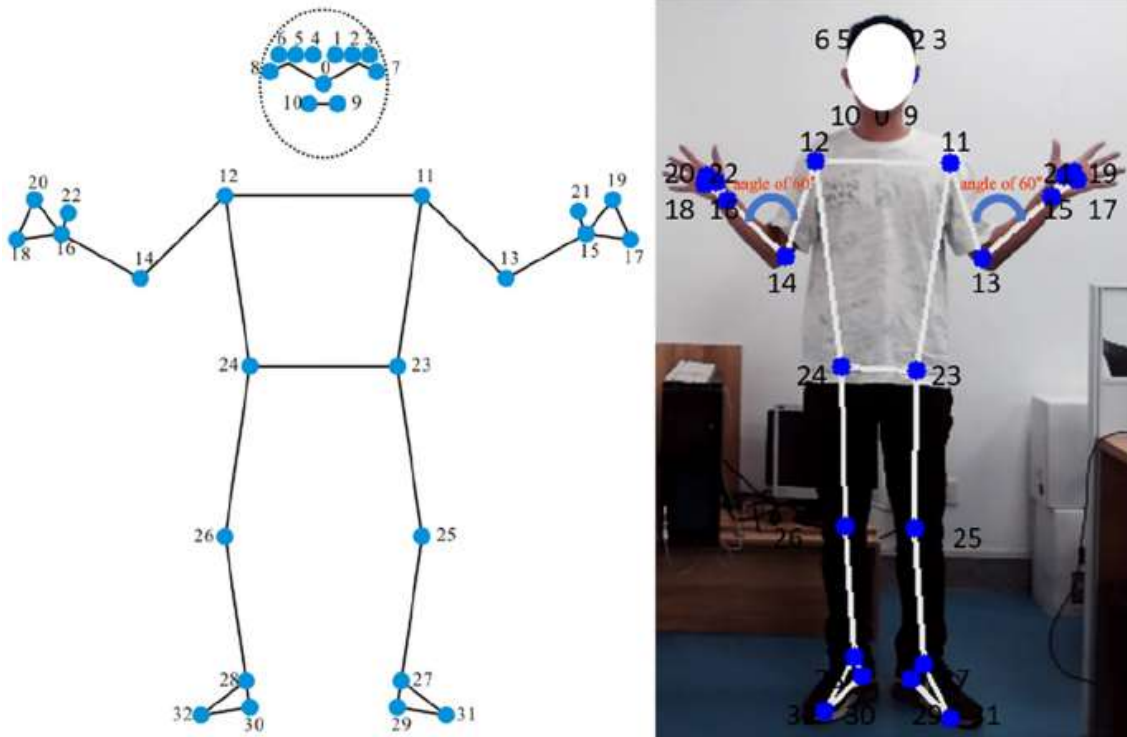


Figure 1: MediaPipe Pose detecting 33 body landmarks used for body measurement estimation.

5. Measurement Parameters Studied

The following measurements were analysed:

Height

Distance between head and feet landmarks.

Shoulder Width

Distance between left and right shoulder landmarks.

Waist Width

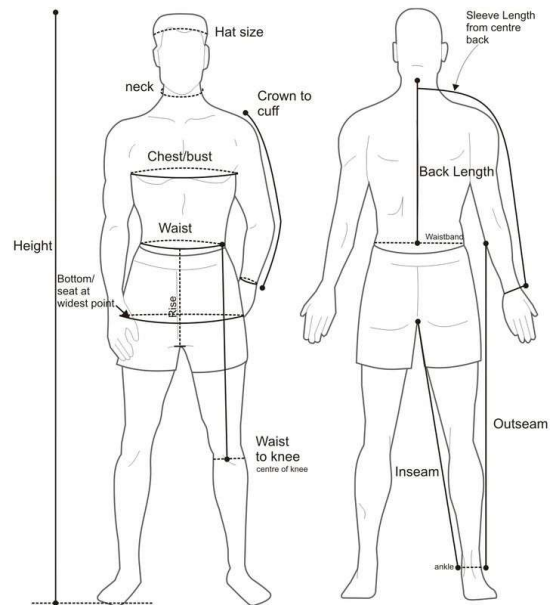
Estimated using torso landmarks.

Hip Width

Distance between left and right hip landmarks.

Arm Length

Distance between shoulder, elbow, and wrist landmarks.



6. AI-Based Body Measurement Estimation Techniques

Landmark-Based Estimation

Landmark-based methods estimate measurements using body joints detected by pose estimation models.

Examples:

- MediaPipe Pose
- OpenPose
- BlazePose

Advantages:

- Fast
- Real-time
- Easy implementation

Limitations:

- Sensitive to camera angle

- Accuracy depends on landmark quality
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Segmentation-Based Estimation

Body segmentation models separate the human body from the background and estimate body dimensions using body contours.

Examples:

- U2Net
- Detectron2
- Selfie Segmentation

Advantages:

- Better body boundary estimation
- Improved waist and hip measurements

Limitations:

- Higher computational cost
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3D Reconstruction-Based Estimation

Advanced systems reconstruct a 3D human body model before extracting measurements.

Examples:

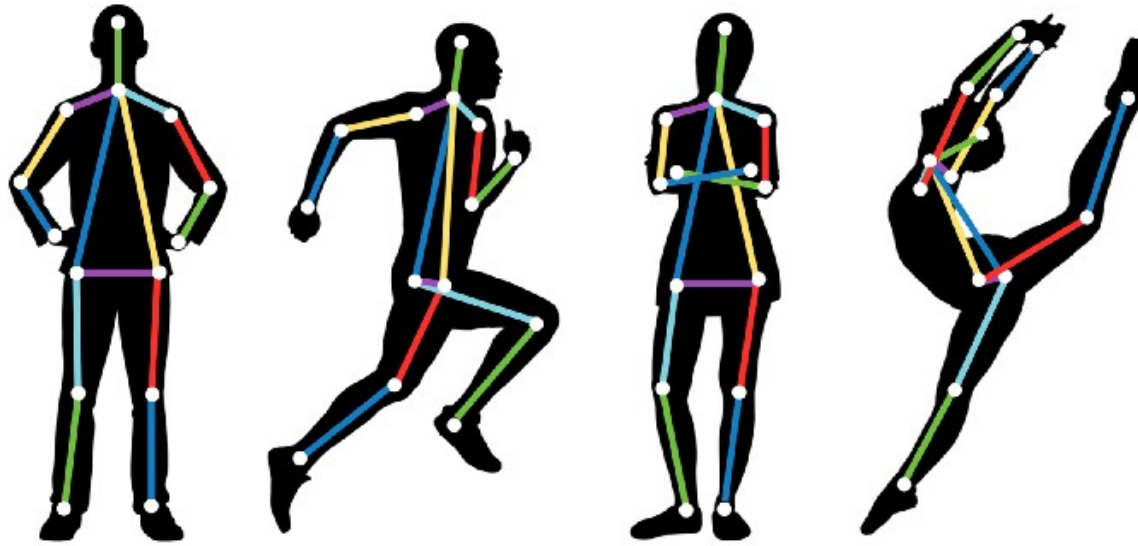
- SMPL
- HMR
- PIFuHD

Advantages:

- Highest accuracy
- Full-body measurements

Limitations:

- Requires powerful hardware



7. Shoulder Width Estimation

Method

Shoulder width is calculated using:

Left Shoulder Landmark

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Right Shoulder Landmark

The pixel distance is converted into approximate physical measurements.

Observations

- Front-facing poses provide the best results.
- Shoulder visibility significantly affects accuracy.

8. Waist and Hip Measurement Estimation

Method

Waist and hip measurements are estimated using torso and hip landmarks.

Body segmentation and landmark localization are combined to improve estimation accuracy.

Observations

- Loose clothing reduces accuracy.
 - Body posture influences measurements.
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9. Arm Length Estimation

Method

Arm length is estimated using:

Shoulder

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Elbow

↓

Wrist

The combined segment lengths provide arm measurement estimates.

Observations

- Straight arms produce better results.
 - Bent arms introduce estimation errors.
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10. Sample Measurement Outputs

Example Output:

Height: 170 cm

Shoulder Width: 42 cm

Waist Width: 34 cm

Hip Width: 38 cm

Arm Length: 60 cm

These values demonstrate how body landmarks can be converted into approximate body dimensions.

11. Accuracy Evaluation

The accuracy of body measurement extraction depends on several factors.

Camera Distance

Subjects positioned too close or too far from the camera may introduce scale errors.

Camera Perspective

Side-view images reduce measurement reliability.

Body Posture

Straight standing poses provide the most accurate results.

Clothing Type

Loose clothing can obscure body boundaries.

Image Resolution

Higher-resolution images improve landmark localization.

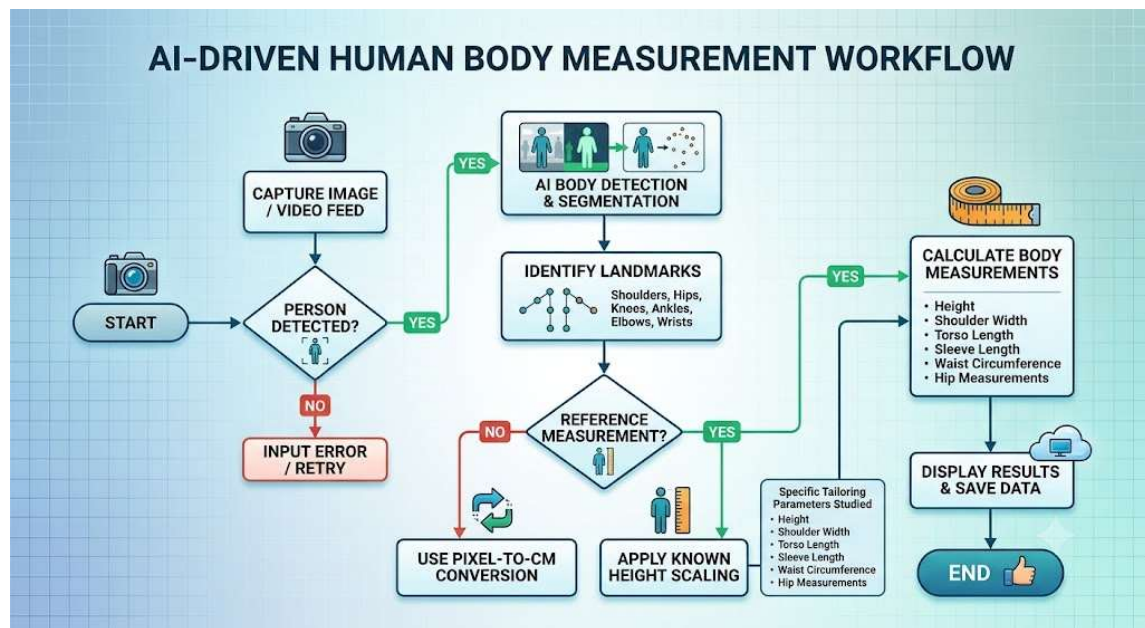
Lighting Conditions

Adequate lighting improves landmark confidence scores.

12. Applications

Body measurement extraction is used in:

- Virtual Try-On systems
- Fashion technology
- Digital tailoring
- Size recommendation systems
- Health monitoring
- Avatar generation



13. Real-World Applications

AI-based body measurement extraction is used in:

Fashion Industry

- Online size recommendation
- Personalized clothing suggestions
- Virtual fitting rooms

Healthcare

- Posture analysis
- Physical rehabilitation
- Movement assessment

Fitness Industry

- Body tracking
- Progress monitoring
- Exercise evaluation

AR/VR Applications

- Avatar creation
- Motion capture
- Virtual garment fitting

E-Commerce

- Reduced return rates
- Improved customer experience
- Personalized shopping

14. Research Findings

The following observations were obtained during the study:

Height Estimation

Most reliable when the full body is visible.

Shoulder Width Estimation

Produces accurate results for front-facing images.

Waist Width Estimation

Improved when segmentation techniques are combined with landmark detection.

Hip Width Estimation

Requires accurate lower-body landmark localization.

Arm Length Estimation

Most accurate when arms remain straight.

Overall Observation

MediaPipe Pose provides a lightweight and effective solution for body measurement extraction, making it suitable for Virtual Try-On applications.

15. Learning Outcomes

During this research, the following concepts were studied:

- Pose estimation
 - Landmark detection
 - Body measurement extraction
 - Computer vision workflows
 - Measurement estimation techniques
 - Virtual Try-On applications
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16. Conclusion

This research explored AI-based body measurement extraction using MediaPipe Pose and OpenCV. The study demonstrated how body landmarks can be utilized to estimate measurements such as height, shoulder width, waist width, hip width, and arm length from a single image.

The analysis showed that measurement accuracy is influenced by factors such as image quality, body posture, camera angle, and lighting conditions. Despite these limitations, landmark-based measurement extraction provides a practical and efficient approach for estimating body dimensions.

The findings highlight the importance of body measurement extraction in applications such as Virtual Try-On systems, digital tailoring, size recommendation platforms, and fashion technology solutions. Overall, the research provided valuable insights into how computer vision and pose estimation techniques can support automated body analysis and personalized user experiences.